



Personalized Asthma Risk Prevention: Synthetic Oversampling for Enhanced Classification on Imbalanced Data

Angelo Alfonso, Linh Le, David Stanko, Mentored by Dr. Wan Bae
Department of Computer Science, Seattle University



Abstract

Individual asthma patient data contains an imbalanced distribution of data samples among different classes, which poses a challenge to machine learning-based asthma risk prediction. To improve the performance of prediction models on imbalanced asthma patient data, we:

1. Investigated existing synthetic minority oversampling technique (SMOTE) variants at the data level
2. Proposed three new minority oversampling techniques
3. Evaluated the effectiveness of the proposed techniques compared to the existing SMOTE variants

The evaluation results showed that the proposed methods are comparable with state-of-the-art oversampling methods.

Introduction/Background

This research focuses on improving the performance of classifiers on imbalanced and small-sized datasets using oversampling techniques, as these issues significantly affect medical datasets in healthcare applications. Although SMOTE variants are commonly used for the class imbalance problem, the effectiveness of SMOTE variants is not well studied. Therefore, we investigated five state-of-the-art SMOTE variants [1–5] and proposed new directions of oversampling methods.

Objectives

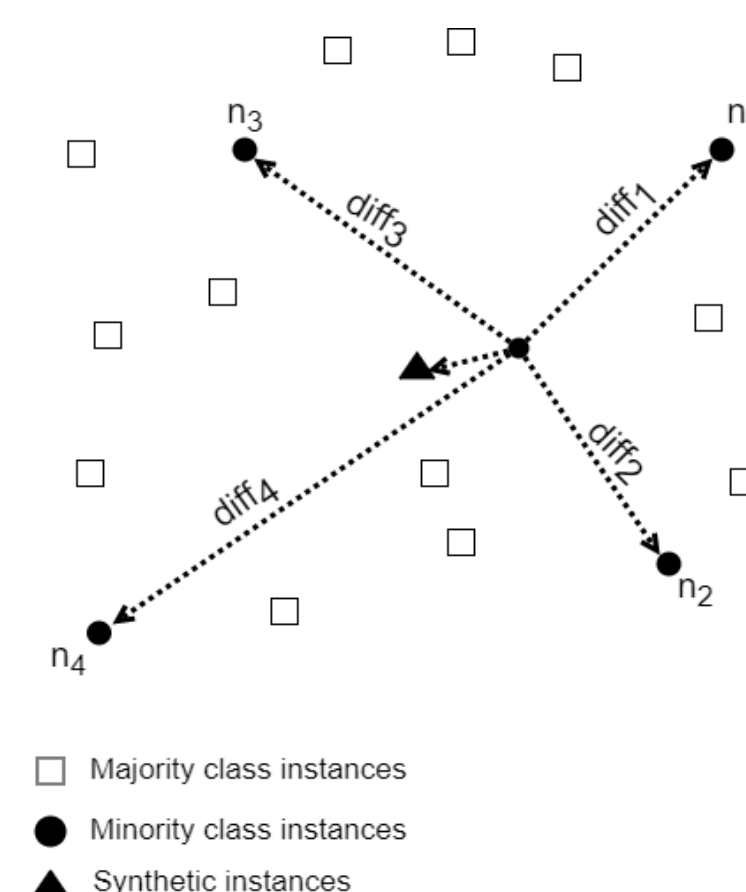
- Develop a framework for evaluating SMOTE variants
- Conduct analysis to understand the effect of each SMOTE variant on each baseline classification model

Methodology

Our focus is to evaluate algorithm performance (particularly sensitivity/recall) on asthma data using 4 classification algorithms and a transfer learning framework.

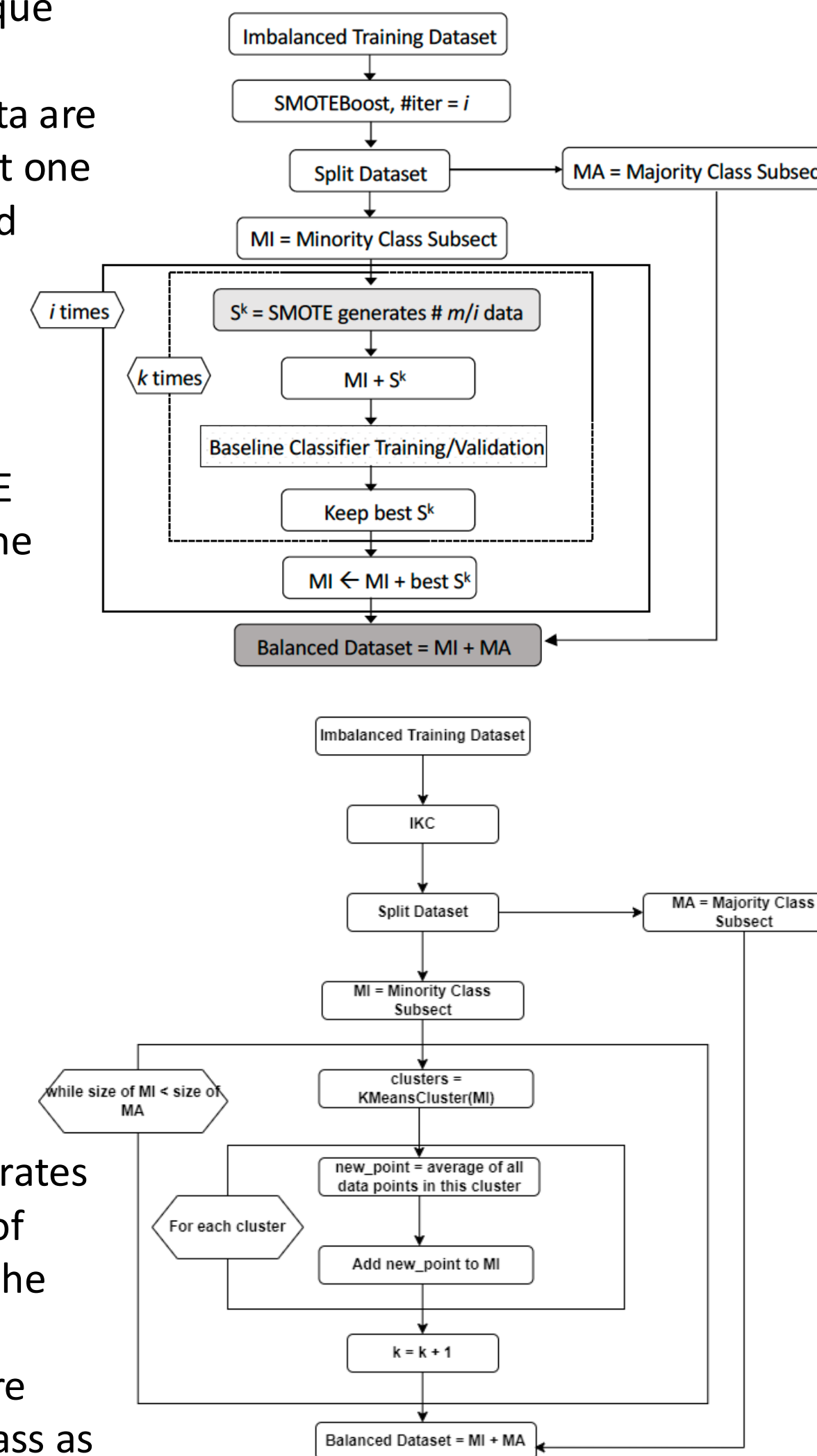
Average Neighbor Vector Oversampling

- ANVO finds the k-nearest neighbors of a minority data point and computes the difference, as a vector, between the point and each neighbor.
- The new data point is the sum of the minority data point and the average vector of those differences.
- Each iteration, we vary the value of k to generate more diverse data.



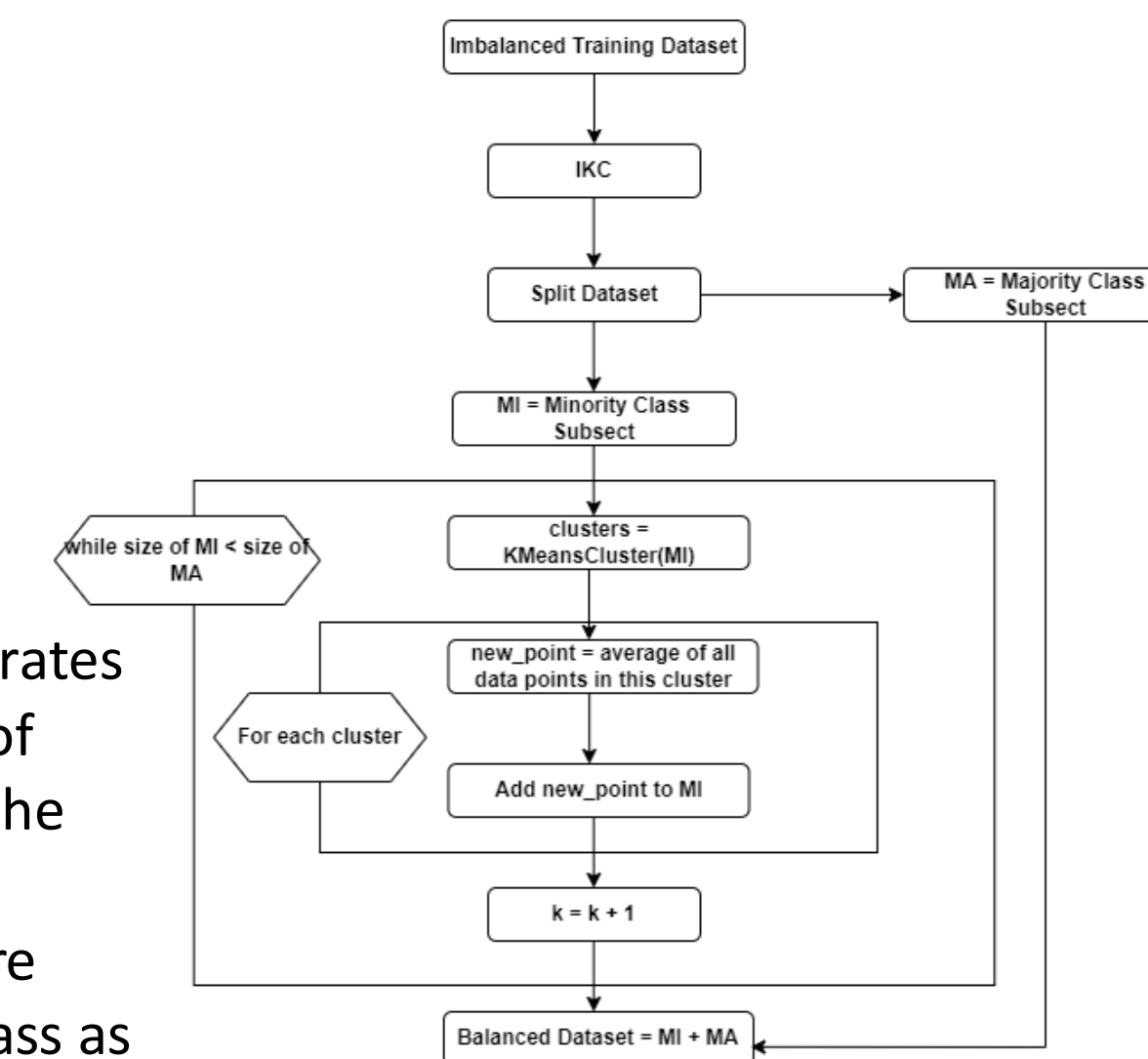
SMOTEBoost and SMOTEBoost with Control Coefficient

- SMOTEBoost adds a boosting technique to the SMOTE algorithm.
- For each iteration, five batches of data are generated using SMOTE, and the best one is chosen by running each augmented dataset through the LR classifier.
- The batch with the best sensitivity is added to the original minority data. SMOTEBoostCC adopts the Control Coefficient (CC) from the SDD-SMOTE algorithm to solve the limitation of the random number value used when synthesizing a new sample.

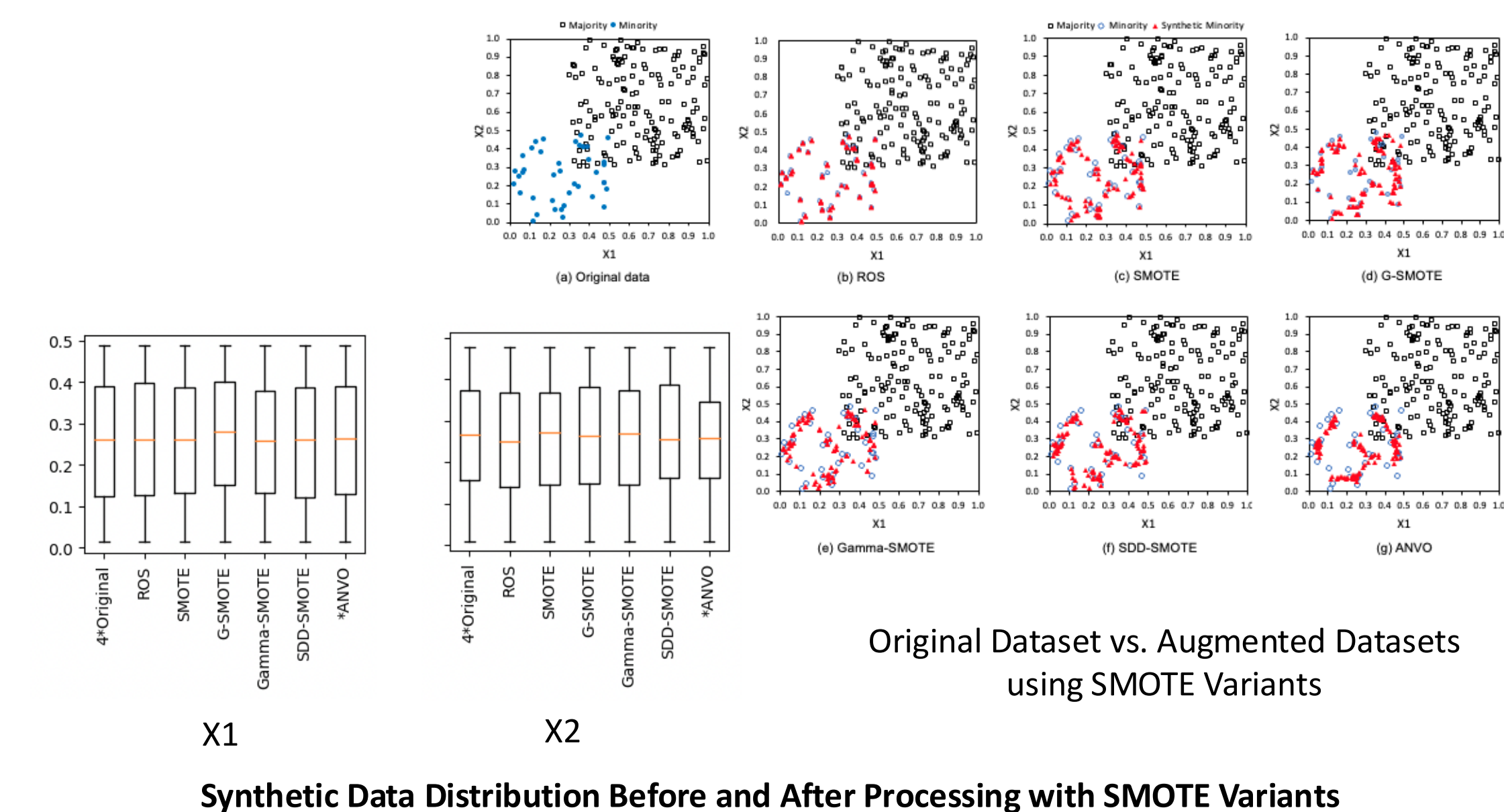


Incremental K-Means Clustering

- IKC uses K-means clustering, where centroids of k clusters are calculated iteratively until the best centroid is discovered.
- IKC utilizes boosting to increase the uniqueness of data points.
- IKC divides the minority class into k clusters (starting with k=2) and generates new points using the average value of each cluster, keeping them close to the original ones.
- For each iteration, generated data are introduced back into the minority class as the k-value increments by one.



Statistical Data Analysis



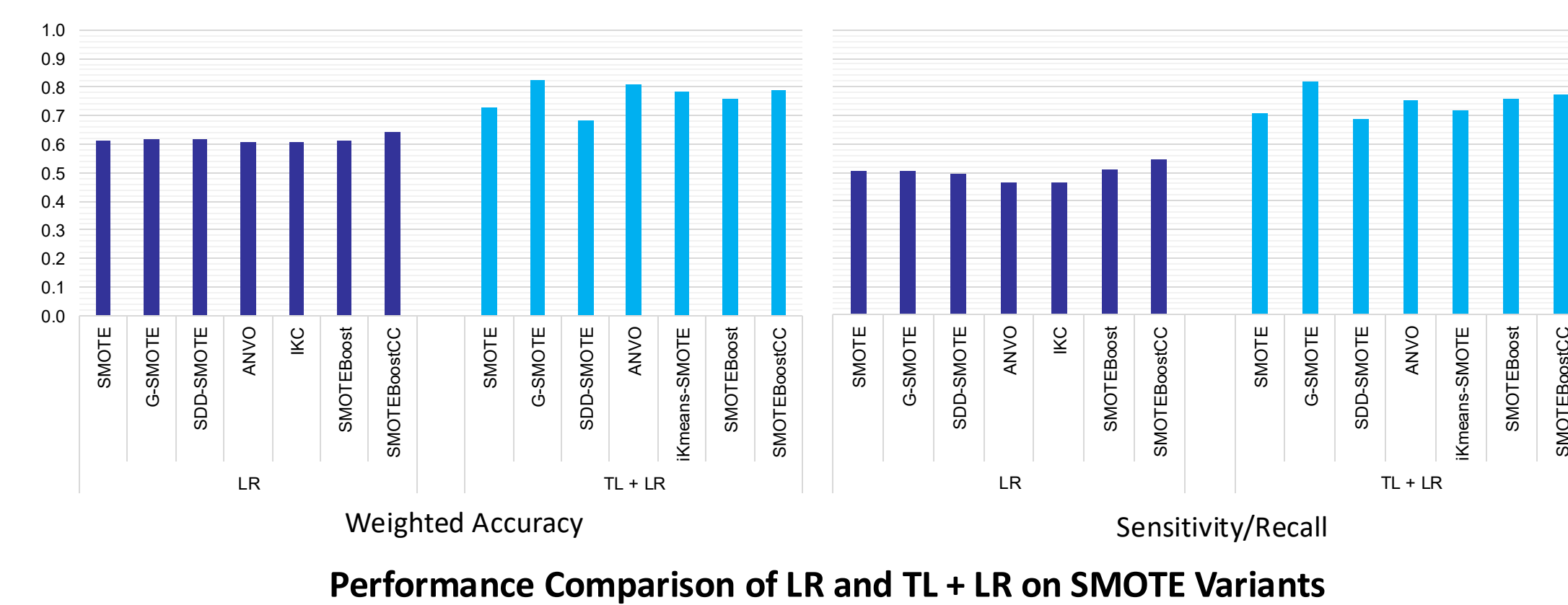
Original Dataset vs. Augmented Datasets using SMOTE Variants

Experimental Results

- We compared the performance of our approaches to that of the five existing approaches using varying sizes of synthetic data and real asthma patients' data.
- Classifiers: Decision Tree (DT), Logistic Regression (LR), k-Nearest Neighbors (KNN), Naïve Bayes (NB), Transfer Learning + Logistic Regression (TL + LR)

Sampling Method	Weighted Acc.	Sensitivity	Precision	F1-Score	Specificity
ORIGINAL	0.5303	0.0762	0.4656	0.4864	0.9844
ROS	0.6088	0.5175	0.5716	0.5606	0.7002
GAMMA	0.6215	0.5295	0.5918	0.5812	0.7134
SMOTE	0.6168	0.5028	0.5933	0.5852	0.7308
SDD-SMOTE	0.6240	0.5390	0.5894	0.5805	0.7089
G-SMOTE	0.6199	0.5060	0.5992	0.5884	0.7338
ANVO	0.6194	0.5073	0.5971	0.5847	0.7315
IKC	0.6213	0.4834	0.5945	0.5892	0.7592
SMOTEBoost	0.6139	0.5115	0.5814	0.5727	0.7162
SMOTEBoostCC	0.6453	0.5447	0.6097	0.6043	0.7459

Averages of Performance on 24 Asthma Patients' Datasets – Logistic Regression



Future Work

- Using stratified sampling to split the data into training and testing sets, instead of random sampling
- Utilizing Generative Adversarial Networks (GAN) as a pipeline for sampling algorithms

Acknowledgements

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